

# Data Understanding Report

## Project Title

Healthcare Analytics and Intelligence Platform

## Dataset Overview

The dataset contains healthcare-related patient records and consists of 55,500 observations and 15 features. The data includes demographic information, medical conditions, hospital details, insurance information, billing amounts, admission details, medications, and test results. The dataset provides a comprehensive view of patient healthcare interactions and is suitable for healthcare analytics, predictive modeling, and business intelligence applications.

## Dataset Structure

- Total Records: 55,500
- Total Features: 15
- Numerical Columns: 3
  - Age
  - Billing Amount
  - Room Number
- Categorical Columns: 12
  - Name
  - Gender
  - Blood Type
  - Medical Condition
  - Date of Admission
  - Doctor
  - Hospital
  - Insurance Provider
  - Admission Type
  - Discharge Date
  - Medication
  - Test Results

## Data Quality Assessment

### Missing Values Analysis

A missing value assessment was performed on all columns. The results indicate that there are no missing values in the dataset.

- Total Missing Values: 0

- Data Completeness: 100%

## **Duplicate Records Analysis**

Duplicate record analysis identified 534 duplicate rows within the dataset.

- Duplicate Records: 534
- Recommendation: Remove duplicate records during the data cleaning phase to improve data quality and model performance.

## **Categorical Data Overview**

Key categorical variables include:

- Gender (2 categories)
- Blood Type (8 categories)
- Medical Condition (6 categories)
- Insurance Provider (5 categories)
- Admission Type (3 categories)
- Medication (5 categories)
- Test Results (3 categories)

These variables provide opportunities for segmentation, statistical analysis, and predictive modeling.

## **Initial Observations**

- The dataset is large enough to support advanced analytics and machine learning applications.
- No missing values were detected, reducing preprocessing complexity.
- A small number of duplicate records were identified and should be addressed.
- Medical conditions, admission types, medications, and test results can be used for classification and clustering tasks.
- Admission and discharge dates can be utilized for trend analysis and time-series forecasting.
- Billing information enables financial and operational healthcare analytics.

## **Potential Analytical Applications**

- Exploratory Data Analysis (EDA)
- SQL-Based Healthcare Analytics
- Statistical Hypothesis Testing
- Patient Segmentation using Clustering
- Admission Type Prediction
- Test Result Prediction
- Time-Series Trend Analysis
- Explainable AI
- Power BI Dashboard Development
- Business Intelligence and Decision Support

## **Conclusion**

The dataset demonstrates high data quality with complete records and sufficient volume for advanced healthcare analytics. After handling duplicate records and performing data preprocessing, the dataset will serve as a strong foundation for building an end-to-end Healthcare Analytics and Intelligence Platform.